6(a)	flow of charge carriers	**B1**
6(b)(i)	*nALe*	**B1**
6(b)(ii)	(t is time taken for electrons to move length L) $I = Q / t$	**B1**
	$I = nALe / t$ **or** $I = nALe / (L / v)$ **or** $I = nAvte / t$ **and** $I = nAve$	**B1**
6(c)(i)	ratio = area at X / area at Y $= [\pi d^2 / 4] / [\pi(0.69d)^2 / 4]$ **or** $d^2 / (0.69d)^2$ **or** $1 / 0.69^2$	**C1**
	= 2.1	**A1**
6(c)(ii)	**1.** $R = \rho L / A$ **or** $R / L \propto 1 / A$	**C1**
	resistance per unit length $= 1.7 \times 10^{-2} \times$ (area at X / area at Y) $= 1.7 \times 10^{-2} \times 2.1$ $= 3.6 \times 10^{-2}\,\Omega\,\text{m}^{-1}$	**A1**
	2. $P = I^2R$ **or** $P = V^2 / R$	**C1**
	$R = 3.6 \times 10^{-2} \times 3.0 \times 10^{-3}$ ($= 1.08 \times 10^{-4}\,\Omega$) $P = 0.50^2 \times 1.08 \times 10^{-4}$ **or** $P = (5.4 \times 10^{-5})^2 / 1.08 \times 10^{-4}$ $= 2.7 \times 10^{-5}$ W	**A1**

Question	Answer	Marks
6(c)(iii)	(cross-sectional area decreases so) resistance increases	**M1**
	($P = I^2R$, so) power increases	**A1**

Question	Answer	Marks